4.1 Introduction

The electric fan, electric washing machine and pumps are essential in domestic for everyday life. In these appliances, electrical energy is converted into mechanical energy. Let us see the appliances in detail.

4.2 Electric fan

According to its utility, the electric fan is classified as,

i. Ceiling fan
ii. Table fan and
iii. Exhaust fan

4.2.1 Ceiling fan

This type of electric fan is fixed in the ceiling of the roof as shown in figure 4.1. In this, the energy conversion is from electrical to mechanical energy and gives cool air inside the room.



Fig. 4.1 **Ceiling fan**

The most important parts of the ceiling fan are

i. Stator
ii. Rotor
iii. Blades
iv. Bearings
v. Down rod

Construction

i. Stator

The stator is provided with insulated silicon steel plate as shown in figure 4.2. Both the starting and running coils are wounded with 90^0 electric degrees.



Fig. 4.2 **Stator**

ii. Rotor



Fig. 4.3 **Rotor**

The rotating part of the squirrel cage rotor type is as shown in figure 4.3. Only in ceiling fan stator is kept inside and rotor is in outside of the fan.

iii. Blade

The blades are made of a steel plate or aluminum plate, as shown in figure 4.4, and are usually fixed with three or four

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